

CHS

Complete Housing Solutions

SYSTEM DOCUMENTATION

Real estate super-app — Nigeria

Document prepared: September 2026

Compiled from the project's live handover records, source code, and database migration history

DOCUMENT CONTROL

Field	Detail
System name	CHS (Complete Housing Solutions)
Document type	System Documentation (compiled reference)
Source materials	CHS_COMPLETE_HANDBOOK_PACKAGE.md, CHS_HANDBOOK_NOTES.md, CHS_PROGRESS_LOG.md, chs-v2 source code (Next.js), backend-v2 SQL migrations (159 files), CHS Feature Catalog (segmented, 193 entries)
Status	Living document — reflects the state of the codebase and database as of the last handover, not necessarily the current live state. Re-verify against the live Supabase project before relying on any specific number or claim below.
Prepared for	Internal engineering handover / onboarding of a new developer or agent

1. INTRODUCTION

This document is a compiled system documentation reference for CHS (Complete Housing Solutions), assembled from the project's own real, verified handover records rather than reconstructed from general assumptions about how such a system might work. Its purpose is to give a new developer, engineer, or technical stakeholder a complete picture of what the system is, how it is built, how its pieces fit together, and what state it is honestly in — including what is proven, what is unfinished, and what has been deliberately built a certain way for reasons that are easy to miss.

Every claim of a feature being "real" or "working" in the source material reflects that it was independently tested against the live system with real data, not merely reviewed in code — a distinction the original project team was careful to maintain throughout, and one this document preserves.

2. SYSTEM OVERVIEW & BUSINESS MODEL

CHS is a real estate super-app built for the Nigerian market (developed and tested primarily around Kaduna State, with Ikeja/Lagos and other cities used in later testing rounds). It covers the full property lifecycle under one platform: buying, selling, renting, leasing, rent-to-own, short-let/hotel/event-venue bookings, property management, a building-materials and services marketplace, and artisan/maintenance job matching.

Core Business Model

CHS never allows money to move directly between two users without first passing through CHS's own wallet system. Every transaction category — sale, rental, shortlet, rent-to-own, marketplace, artisan labour — carries its own specific commission percentage, charged to one or both parties. This escrow-and-commission pattern is the single most important architectural fact about the system, and it is enforced consistently at the database function level (see Section 7).

Deployment Footprint

Frontend hosted on Netlify, backend/database/auth/storage on Supabase, source control on GitHub. Deployment is currently a manual process: the working developer builds and packages the app, and updates are pushed to GitHub in batches, which triggers a Netlify rebuild.

3. USER ROLES

CHS supports the following distinct roles. A single person may hold more than one role on a single account (e.g. Owner and Tenant simultaneously) via a "Link Account" flow, rather than maintaining separate registrations.

Role	Summary
Buyer	Browses and purchases property (outright sale or rent-to-own).
Tenant	Rents property; pays rent and manages tenancy.
Owner	Lists property for sale/rent; manages offers, applications, and maintenance.
Agent	Manages listings and clients on behalf of owners; earns commission; can manage a team.
Property Manager	Manages estates and tenancies on behalf of owners.
Commercial Developer	Registers and manages larger-scale development listings.
Vendor	Sells building materials/products via the Marketplace.
Artisan	Offers maintenance/repair labour; quoted and paid through the platform.

Role	Summary
Host	Lists property for short-let, hire, or event-venue booking.
Guest	Books short-let, hire, or event-venue services.
Staff/Employee	Team member under an Agent or Manager account.
Admin	Platform oversight — organised into sub-admin "domains" (owner_buyer_tenant, agent_manager, vendor_artisan, etc.) plus a super-admin role with full access.

4. SYSTEM ARCHITECTURE

Frontend

Next.js 16.2.12 using the App Router, React 19.2.4, TypeScript 5, Tailwind CSS 4. Deployed to Netlify. An alternative Cloudflare Workers build path exists (@opennextjs/cloudflare, wrangler) but is configured only, not the live deployment target.

Backend

Supabase (PostgreSQL + Auth + Storage + Realtime). The overwhelming majority of business logic lives in PostgreSQL functions (SECURITY DEFINER, called via supabase.rpc(...) from the client) rather than in Next.js API routes — this application has no real Next.js API routes at all. Every money-moving or state-changing operation is implemented as a single atomic Postgres function.

Request/Feature Pattern

1. A Postgres function in backend-v2/ validates the caller via auth.uid(), moves money through the wallets/wallet_transactions tables where applicable, inserts/updates the relevant tables, and calls notify_user(...) for any party who needs to be informed.
2. A Next.js page or component invokes this function via supabase.rpc("function_name", { p_param: value })).
3. Real-time or on-load queries read the resulting state directly from the relevant tables, often via a Supabase join (e.g. properties(title, ...)).

Authentication Pattern

Users register with a phone number. Internally, this becomes a synthetic email (chuser{phone}@chsplatform.app) for Supabase Auth, while the phone number remains the human-facing identifier everywhere in the UI. PIN-based login uses this synthetic email plus a 6-digit PIN as the password. Biometric login uses WebAuthn (@simplewebauthn/browser and @simplewebauthn/server), backed by four Supabase Edge Functions. This pattern is intentional — Supabase Auth requires an email, while CHS's real identity system is phone-based — and should not be "simplified" to real emails without understanding every downstream place phone-based lookup is relied upon.

UI Theming Pattern

Every dashboard's root element carries a zone-{role} class, and each role's header uses the CSS custom property --zone-accent (never a hardcoded colour) for its top bar. This is what gives each role's dashboard a visually distinct identity. Hardcoding a colour on a new dashboard header has previously caused a reported bug where a new role's dashboard looked identical to another's.

5. TECHNOLOGY STACK

Layer	Technology
Frontend framework	Next.js 16.2.12 (App Router), React 19.2.4, TypeScript 5
Styling	Tailwind CSS 4 — no component library; all UI hand-built
State management	Plain React state + Supabase queries — no Redux/Zustand
Backend / database	Supabase (PostgreSQL, Auth, Storage, Realtime)
Serverless functions	Supabase Edge Functions (WebAuthn ceremony — 4 functions)
Payments	Paystack (@paystack/inline-js) — currently test keys only

Layer	Technology
Biometric auth	@simplewebauthn/browser, @simplewebauthn/server
PDF generation	jspdf (client-side, where used)
Hosting	Netlify (live/current); Cloudflare Workers via wrangler + @opennextjs/cloudflare (configured, not live)
SMS	Termii — built, provider sender-ID approval pending
Email	Resend — built, provider approval pending

6. DATABASE ARCHITECTURE & MIGRATION HISTORY

The database schema and all business logic functions live in backend-v2/, as 159 sequentially numbered, self-documenting SQL migration files (from 1_... through 222_..., with some numbers skipped where a migration was abandoned before being applied). The legacy backend/ folder (10 files) represents an earlier, pre-v2 backend attempt, superseded and not used by the live application. Migrations group into the following honest phases:

Range	Phase
1–50 (approx.)	Core schema — profiles, properties, wallets, offers, tenancies, original registration/KYC system, basic commission logic.
50s–90s	Bug-fix rounds following early client testing — dispute rulings with no real consequence, agent referral payouts that never fired, management termination that never worked, developer approval with no real consequence, maintenance reserve never wired up. Filenames are prefixed fix_ and name the bug directly.
90–150	Feature rounds — Rent-to-Own, shortlet messaging/escalation, sale legal-document + escrow flow, agent full-authority management, precommit message moderation, estate management, team subscriptions, KYC fixes, property reference numbers, PIN reset, FAQ, per-role colour themes.
150–195	Team/business tools for agents and managers, Host/Guest/Developer as separate roles, complete shortlet/hire rebuild (request-to-book, ratings, cancellation policy, security deposits), receipt/voucher system and redesign, tiered artisan commission.
196–222 (most recent)	Complete marketplace rebuild (admin-mediated messaging, escrow payment, direct-buy), sister marketplace link, rental-application rework with independent guarantor verification, Event Centre capacity tiers and facilities, payment-regression fixes (pay_rent, pay_rent_to_own_installment), stale-commission early-warning alert, notification/real-time overhaul.

Guidance for any future work: read migration filenames in numeric order — each carries its own top comment explaining the real problem it fixes or the feature it adds. Never renumber or delete an existing migration file; add new ones sequentially. Before making any schema change, confirm the live Supabase migration count still matches the highest number recorded here (222 as of this document).

Edge Functions

Four Supabase Edge Functions implement the complete WebAuthn ceremony for biometric login: webauthn-register-options.ts, webauthn-register-verify.ts, webauthn-auth-options.ts, webauthn-auth-verify.ts.

7. KEY DESIGN DECISIONS (things a new developer might otherwise "fix" incorrectly)

1. Every payment computes and charges its own commission within the same atomic function call — never as a separate step a different function is trusted to remember later. This was violated twice (`pay_rent`, `pay_rent_to_own_installment`) and caused a real client-facing bug (money appearing to vanish, no breakdown shown) before being caught and fixed. `pay_for_property` (Sale) was always correct and is the reference pattern to copy.
2. Guarantor details on a rental application must never be entered by anyone but the guarantor. The applicant provides only a name and phone number. This was deliberately split apart after a direct client concern about consent and fraud risk, and should not be simplified back into a single form.
3. The marketplace intentionally has two separate purchase paths: a moderated quote-request negotiation, and a direct "Buy Now" for fixed-price items. Both use the same 6%/4% commission split and the same escrow mechanism — they should not be merged, nor should one be treated as legacy.
4. `transaction_commissions.status` defaults to 'pending' and must be explicitly set to 'paid' the moment money actually moves. A stale-commission alert (Admin → Finance) exists specifically to catch any future function that inserts a commission row and forgets this step.
5. Contact-info blocking (`detect_offplatform_contact`) is a shared, reusable function, used identically across property-offer negotiation and the marketplace. Any new real-time messaging feature between two parties who should not exchange direct contact info should reuse this function rather than reinventing detection logic.
6. Notification links matter as much as the notification itself. `notify_user(p_user_id, p_title, p_body, p_link)` — the 4th parameter (link) has repeatedly been forgotten. A notification without a working link to the relevant page is treated as a reportable bug, not a nice-to-have.
7. Every dashboard's root element carries a `zone-{role}` class, and each role's header uses `bg-[var(--zone-accent)]` rather than a hardcoded colour. This is what makes each role's dashboard visually distinct.
8. Document URLs must never be trusted long-term. `getFreshDocumentUrl()` in `lib/storage.ts` generates a fresh signed URL at the moment a document is actually viewed, because signing-key rotation silently breaks previously stored URLs with no visible error.
9. The synthetic-email auth pattern (`chuser{phone}@chsplatform.app`) is intentional, not a hack to clean up — see Section 4, Authentication Pattern.

8. KNOWN ISSUES & LIMITATIONS

Confirmed, fixed, and proven (documented so they are not re-discovered)

- `pay_rent` and `pay_rent_to_own_installment` silently omitted the payer's own commission from the amount actually charged. Fixed and tested with real numbers (migrations 215–218).
- `submit_rental_application` never notified the property owner of a new application. Fixed (migration 220).
- The notifications table was never added to Supabase's real-time publication, meaning no live pop-up could ever have worked regardless of frontend code. Fixed (migration 222).
- `notify_admins_by_domain` never accepted a link parameter at all. Fixed to match `notify_user`'s signature.
- 6 of 10 marketplace vendor categories (including Building Materials) had no referral fee configured, and the schema blocked one from ever being added. Fixed, then the model was replaced entirely.

Confirmed but not yet fixed — genuinely open items

- No real frontend screen exists for a Rent-to-Own buyer to make an installment payment. The backend function (`pay_rent_to_own_installment`) is correct and tested directly via SQL, but nothing in app/ currently calls it.
- The Saved/Favorite feature was reported by the client as "not working," but direct testing against the live database shows it functions correctly end-to-end. Likely a discoverability issue (the heart icon isn't clearly labelled), not a functional bug — confirm with the client before rebuilding.
- One open Supabase security advisory: `public.public_profiles`, a SECURITY DEFINER view, flagged ERROR-level by Supabase's own advisor. Not yet investigated or resolved.

Structural gaps

- No automated test suite exists anywhere in the project (no Jest/Playwright/Cypress). All verification has been manual — direct Postgres function calls with real test data, checking before/after wallet balances, plus `npm run lint` and `npm run build` for frontend changes.
- No CI/CD pipeline. Deployment is manual: the working developer zips the repo, the client pushes to GitHub (historically in batches), which triggers a Netlify build.

9. EXTERNAL SERVICES & INTEGRATIONS

Service	Used for	Status
Supabase	Entire backend — database, auth, storage, real-time, Edge Functions	Live, active (project ref: havwhdgjqgtxtqlkqfm)
Netlify	Frontend hosting/deployment	Live, active
GitHub	Source control	Live, active — manual push of zip updates
Paystack	Wallet funding / real money movement	Test keys only — real payments not yet live
Termii	SMS (OTPs, rent reminders, etc.)	Built; sender-ID approval pending
Resend	Transactional email	Built; provider approval pending
Cloudflare (wrangler)	Alternative deploy target	Configured, not the live deployment

No other third-party APIs are in use, confirmed by codebase search: no Google Maps, no third-party NIN verification API, no third-party face-liveness service. NIN uniqueness is checked only within CHS's own database, and ID verification is a photo reviewed manually by CHS staff.

10. SECURITY CONSIDERATIONS

- Row-Level Security (RLS) policies gate access throughout the database, layered on top of the SECURITY DEFINER function pattern described in Section 4.
- Private, sensitive documents are served exclusively via freshly generated signed URLs (`getFreshDocumentUrl()`) rather than stored URLs, to survive signing-key rotation without silent breakage.
- Contact-information leakage between negotiating parties is actively detected and blocked (`detect_offplatform_contact`), reused consistently across property offers and the marketplace.
- One open Supabase advisory (ERROR level) on the `public.public_profiles` SECURITY DEFINER view remains unresolved as of this document and should be prioritised.
- Biometric login uses the standard WebAuthn ceremony rather than a custom implementation.
- No API keys, passwords, or service-role secrets are recorded in this document or any of its source material — all remain known only to the account owner.

11. ENVIRONMENT & DEPLOYMENT

Local development setup

cd chs-v2 && npm install

Create chs-v2/.env.local (never commit) with:

NEXT_PUBLIC_SUPABASE_URL=<the real Supabase project URL>

NEXT_PUBLIC_SUPABASE_ANON_KEY=<the real Supabase anon/public key>

Then: npm run dev (local server) · npm run build (must be run before considering any change complete) · npm run lint (should show 0 errors).

Database migrations

Add a new, sequentially numbered .sql file to backend-v2/ (check the highest existing number first) and apply it directly to the live Supabase project. Never renumber or delete an existing migration file.

Deployment

Frontend deploys to Netlify from GitHub. There is no CI/CD pipeline — pushes are manual and batched.

12. RECENT CHANGE HISTORY (most recent work, in order)

1. Receipt/payment voucher redesign (branded document).
2. Artisan commission corrected to tiered 6%/8.5%.
3. Complete marketplace rebuild — admin-mediated messaging, escrow payment, direct-buy path.
4. Sister marketplace link added (placeholder URL — needs real address).
5. Rental application rework — full applicant detail shown to owner.
6. Owner dashboard price-display fix.
7. Event Centre capacity tiers and priced facilities (verified against the legacy prototype file).
8. Independent guarantor verification — the guarantor completes their own separate step.
9. Payment regression found and fixed — pay_rent and pay_rent_to_own_installment both silently omitted commission.
10. Stale-commission early-warning alert added for admin.
11. Notifications made genuinely clickable and genuinely real-time.
12. New dedicated tabs: My Applications, Recent Applications (owner), Platform Earnings (admin); duplicate Rent Savings widget removed from homepage.

13. PENDING WORK (prioritised)

1. Build the Rent-to-Own payment screen. Backend is complete and proven; this is frontend-only work (est. 1–2 hours) — a page showing the buyer's active agreement(s), ownership percentage so far, and a payment button calling pay_rent_to_own_installment.

2. Update the sister marketplace URL from its placeholder (currently a fake address) to the real Unify Market Central address, via Admin → Overview → Edit Real Contact Details. This is an admin action, not a code change.
3. Confirm the Saved/Favorite discoverability question directly with the client before touching that code further.
4. Complete SMS (Termii) and email (Resend) provider approval, then wire the already-built guarantor-confirmation link (and other notifications) to send automatically.
5. Move Paystack off test keys once business verification with Paystack is complete — this unblocks real money entering the system.
6. Investigate and resolve the open Supabase security advisory (public_profiles view).
7. Clean up or clearly mark as dead the two known-inert legacy data points: the old flat artisan commission setting and the pre-rebuild marketplace referral-fee tables.
8. Maintain the established discipline: whenever any payment flow is touched, walk it end-to-end with real data and a real resulting receipt before considering it done.

APPENDIX A — COMPLETE ROUTE MAP (chs-v2/app/)

Every application route and its purpose. Folder path is the real URL route; every file is page.tsx unless noted.

Route	Purpose
/ (page.tsx)	Homepage — property browsing, search, filters. Server component wrapping HomePageClient.
/about	Real About Us content.
/accept-terms	Real, legally-gated T&C acceptance (checkbox only enables after genuine scroll-to-bottom).
/admin	**The single largest file in the app** — the complete admin dashboard, all tabs (overview, analytics, finance, trace, sale approvals, liveness, registrations, applications, properties, disputes, feedback, engage, vendors, referrals, faults, artisans, inspections, developers, tenant register oversight, shortlet/hire deposits, marketplace moderation, platform earnings).
/admin/concierge	Admin side of the "Talk to an Agent" concierge request feature.
/admin/feature-catalog	Real, in-app, always-current rendering of the same feature catalog data used to build the PDF/XLSX.
/admin-approval-pending	Waiting screen for a sub-admin after a correct password, before super-admin grants real access.
/agent	Agent dashboard — listings, team management, commission settings.
/agent/property-register, /agent/tenant-register	Real, dedicated registers for an agent's own managed properties/tenants.
/analytics/[id]	Real per-property analytics (views, inquiries, etc.) for an owner/agent.
/artisan	Artisan dashboard — job quotations, assigned jobs, payments received.
/become-artisan, /become-vendor	Real registration flows for these two categories.
/blog	Real blog/content listing.
/choose-category	Entry point for picking which role/category to register or act under.
/concierge	User-facing "Talk to an Agent" request form.
/condition-report/[tenancyId]	Real move-in/move-out condition report for a tenancy.
/construction-roadmap	Real construction cost/permit roadmap tool, with the sister-marketplace link.
/contact	Real, admin-editable contact details page.
/developer	Commercial Developer dashboard.
/edit-listing/[id]	Real property edit form.
/engage-chs	The "Engage CHS" professional services (Property Management, Sale Negotiation, etc.) request and messaging flow.
/event-pricing/[propertyId]	**New.** Host-side real capacity-tier and facilities-pricing management for Event Centre listings.
/expenses	Agent/Manager expense tracking.
/faq	Real FAQ content, written from CHS's actual T&C.
/forgot-pin	Real PIN reset via phone + NIN verification.
/guarantor-confirm/[token]	**New.** The real, independent, no-login-required page a rental guarantor uses to confirm their own details and consent.
/guest	Guest (shortlet/hotel) dashboard.

Route	Purpose
/guide	In-app Users Guide.
/host	Host (shortlet/hire/event venue owner) dashboard — booking requests, event-day add-on requests, payouts.
/invite/[token]	Real invitation-link handling (e.g., team member invites).
/link-account	Adding a second role to an existing account without re-registering.
/list-property	The real property listing form, branching by category.
/login	Real phone+PIN and biometric login, with real per-role redirect (see roleToPath maps — buyer now correctly goes to /my-offers, not /).
/manager, /manager/estates, /manager/estates/[id]	Property Manager dashboard and Estate Management tools.
/market-demand	Real "demand registry" — buyers/tenants register what they're looking for.
/marketplace	The real, rebuilt marketplace — browse, sort by price, request a quote, or Buy Now.
/my-applications	**New.** Buyer/tenant's permanent queue of every real rental application and purchase offer.
/my-bookings	Guest's shortlet/hire booking history.
/my-offers	Buyer's real purchase offers — also the real post-login landing page for the Buyer role.
/my-quote-requests	Buyer's real marketplace quote requests and direct orders, with Accept & Pay.
/my-receipts	**General, real transaction history** for any user — every real wallet_transactions row, linking to the real receipt/voucher page.
/owner	Owner dashboard — the second-largest file in the app; properties, rental applications (with full real applicant detail), offers, maintenance, everything an owner manages.
/owner-applications	**New.** Owner's dedicated, permanent view of every real application/offer received across every property.
/profile	Real profile management, role switching.
/promote/[id]	Real listing-promotion purchase flow (credits, fixed tiers, subscriptions).
/property/[id]	Real property detail page — the main "apply/offer/book" entry point.
/receipt/[reference]	**The real, branded receipt/payment-voucher document** , auto-labeled by viewer role.
/register	Real account registration.
/saved	Real saved-properties list (confirmed working directly against the database).
/staff	Staff/employee dashboard (for agent/manager team members).
/tenant	Tenant dashboard — tenancies, rent payment (with real commission breakdown shown before paying), notices.
/terms	The real, full Terms & Conditions content.
/urgent-sale/[id]	Real Urgent & Emergency Sale listing detail.
/vendor	Marketplace vendor dashboard — products, quote requests, direct orders.
/wallet	Real wallet management — fund, withdraw, view all balances.
layout.tsx, not-found.tsx, manifest.ts, robots.ts, sitemap.ts	Real Next.js framework files — root layout (loads fonts, NotificationBell, etc.), 404 page, PWA manifest, SEO files.

APPENDIX B — SHARED COMPONENT INVENTORY (58 files)

Auth / security

AccountStatusGate, BiometricLogin, BiometricSetup, IdentityVerificationGate, LivenessCheck, BankAccountSecurity

Navigation / layout

BottomNav, ChsLogo, RoleBadge, SplashScreen, ThemeToggle, NotificationBell (real-time, clickable, toast-enabled), GuidePrompt, TestModeBanner

Property browsing / actions

PropertyCard, PropertySearch, PropertyActions (hub of buy/rent/book buttons), SaveButton, ShareButton, InterestButton, CurrencyInput, CurrencyReference

Booking forms

ShortletBookingForm, HireBookingForm (includes Event Centre tier/facilities UI), InspectionBookingForm, HostBookingDecision, ShortletCheckInOut, ShortletRating, CancelBookingButton, HouseRulesAcknowledgment, HouseRulesUpload

Applications / forms

RentalApplicationForm (split guarantor flow), IssueNoticeForm, RequestTermination, RaiseDisputeForm, RateArtisanForm, PostQuotationJob, DemandRegistryForm

Messaging

MessageThread, OfferMessageThread, ShortletMessageThread, OwnerAdminMessageThread, EngageChatThread — several with admin-moderated contact-info blocking

Marketplace

MarketplaceClient

Documents / content

DocumentViewLink (fresh signed URLs), TermsContent, FaqContent, GuideContent, EngageDocuments, MediaRequests

Miscellaneous

CommunityFeedback, ComprehensionCheck, DiasporaMode, TransactionCommissions, WalletQuickView, ServiceTncGate, ServiceWorkerRegistration, HomePageClient

APPENDIX C — FUNCTIONAL FEATURE CATALOG BY ROLE (193 entries)

The complete functional specification, organised exactly as maintained in the project's own live Feature Catalog. Each entry lists what the feature is, how it works, and where it is found in both the user-facing app and the admin panel.

1. Platform-Wide (Every Role)

Feature	How it works	User location	Admin location
Registration with role selection	Buyer, Tenant, Property Owner, Agent, Property Manager, or Commercial Developer — with role-specific extra fields (e.g. professional certificate upload for Managers).	Register page	New accounts land in Admin → Registrations, pending approval
NIN uniqueness check	A real National Identification Number can only ever be tied to one CHS account.	Register page	N/A — enforced at registration
Multi-role linking	One real identity, one wallet, one login can hold several roles at once (e.g. Owner + Tenant) via "Link Account" rather than a second registration.	Link Account page — every role	N/A — self-service
Biometric login (Face ID / fingerprint)	Real WebAuthn — uses the device's own secure hardware; CHS never sees or stores the actual biometric data.	Login page + Profile setup — every role	N/A — self-service
Valid ID + liveness verification	Real document upload plus a genuine, real-time photo capture (never an uploaded photo) reviewed by CHS staff.	Registration / Profile — every role	Admin → Face Verification (Registration & Setup domain)
Editable profile with re-verification trigger	Name, phone, address, and state are genuinely editable. A real name change automatically resets ID/liveness verification.	Profile page — every role	Real notification to Super Admin on any name change
Wallet (main + rent savings + maintenance reserve)	Three real sub-balances per user — a main spending balance plus two dedicated savings pots for rent renewal and maintenance costs.	Wallet page — every role	Admin → Finance — search any user's real balances, freeze/unfreeze
Wallet funding via Paystack	Real, secure card/bank/USSD payment; balance updates automatically once payment genuinely completes.	Wallet page — every role	N/A — self-service
Wallet withdrawal to bank	Only one linked bank account per wallet, for security. Real Paystack transfer to the verified account.	Wallet page — every role	N/A — self-service, admin sees real withdrawal_requests records
Wallet-to-wallet transfer	Send funds directly to another CHS user by phone or email — real-time lookup and instant transfer.	Wallet page — every role	N/A — self-service
Bank account linking with real-time verification	200+ real Nigerian banks with search; the account number is checked against the real bank network before it can ever be linked. Real 48-hour security window on any change.	Wallet → Link bank account — every role	N/A — self-service, real-time
Notification preferences	Independent real toggles for offers, messages, and marketing notifications.	Profile page — every role	N/A — self-service
Diaspora Mode	A real setting for CHS users managing property or transactions from outside Nigeria.	Profile page — every role	N/A — self-service
Terms & Conditions scroll-to-accept gate	A real, meaningful acceptance — the checkbox only unlocks after genuinely scrolling to the bottom. Required before any dashboard is reachable.	First login — every role	N/A — one-time gate
Role-specific first-dashboard guide + persistent Users Guide	A short popup on first dashboard visit, plus a full, always-accessible reference page.	First login popup + More menu → Users Guide	N/A — reference

Feature	How it works	User location	Admin location
Dark mode	A real, complete theme system across the whole app, with an app-wide fix ensuring form text stays readable in both modes.	Every page — every role	N/A — visual
PWA install + splash screen	Installable as a real app on the home screen, with a proper full-screen splash and app icon.	Browser install prompt	N/A — visual
Guest — its own real role	A genuine, separate role from Buyer, for booking a shortlet apartment or a hotel/lodge/event centre/car park stay — its own heading at registration and login, not merged into another role.	Register / Login page — every role	N/A — self-service
Account approval gate	A real, enforced rule: a newly registered account cannot use anything beyond the public homepage until CHS staff approve it — previously unenforced despite existing in the database.	Every page — every role	Admin → Registrations
Account suspension & appeal	Real admin tools to suspend a suspicious account with a documented reason, and a genuine appeal channel that stays reachable even while suspended.	Suspended-account screen — every role	Admin → Overview, Suspend a Real Account
Self-service account deactivation	A real, reversible way to temporarily deactivate your own account — nothing deleted, log back in any time to reactivate.	Profile page — every role	N/A — self-service
Automatic image compression	Every property photo and profile picture is resized and re-compressed in the browser before upload — no action required, real storage savings confirmed by direct testing.	Every photo upload — every role	N/A — automatic
Digital receipts	A real, printable receipt (Print/Save as PDF) for any transaction — rent, remittance, commission, sale — using the same reference number already generated for it.	My Receipts page — every role	N/A — self-service

2. Buyer & Tenant

Feature	How it works	User location	Admin location
Location + price search	Filter by state, LGA, area, property type, minimum bedrooms, and a real minimum/maximum price range.	Homepage search — Buyer / Tenant	N/A — self-service
Rent / Sale / Shortlet tabs	Real filtering of the whole catalog by purpose.	Homepage — Buyer / Tenant	N/A
Saved Properties & Saved Searches	Save individual listings, or save a whole search to be matched automatically as new listings arrive.	Saved page — Buyer / Tenant	N/A — self-service
Making an offer (sale)	Propose a real price directly to the owner; goes through a real CHS Sale Approvals checkpoint before anything moves to escrow.	Property detail page — Buyer	Admin → Sale Approvals (Owner/Buyer/Tenant domain)
Rental application	A real application, screened by CHS staff before the owner ever sees it.	Property detail page — Tenant	Admin → Applications (Owner/Buyer/Tenant domain)
Inspection booking + transport fee	Real fee calculated per-person based on distance; a genuine 12-hour minimum notice enforced by the database itself.	Property detail page — Buyer / Tenant	Admin → Inspections (Owner/Buyer/Tenant domain)
Buyer/Tenant Readiness Score	A real questionnaire (timeline, funds ready, decision-maker) plus verification and inspection-attendance history — shown to owners, never a fee or gate.	Inspection booking form — Buyer / Tenant	N/A — computed live
Shortlet booking	Paid instantly through the wallet, held in real escrow until check-in is confirmed. Real guest verification required.	Property detail page — Buyer / Tenant	N/A — self-service
Rent-to-Own	Where offered, a real monthly amount counting toward eventual ownership, shown directly on the listing.	Property detail page — Buyer	N/A — set by owner at listing time
House Rules acknowledgment	A real, documented agreement a tenant acknowledges as part of their tenancy.	Tenant dashboard, per tenancy	N/A — set by owner
Property Condition Report	Documented and photographed at move-in — real evidence-based liability instead of a caution-fee guess.	Condition Report page — Tenant	N/A — self-service
Formal Notices	Real formal communications (e.g. rent reminders, lease notices) from owner/CHS, viewable and expandable.	Tenant dashboard	N/A — sent by owner/CHS
Owner–tenant message thread	A real, direct messaging channel per tenancy.	Tenant dashboard, per tenancy	N/A — self-service
Raise a Dispute	A real, two-sided dispute process — CHS's internal resolution is the first step before arbitration or courts.	Tenant dashboard + Owner dashboard	Admin → Disputes (Customer Care domain)
Talk to an Agent (Concierge)	Describe a property need in free text or voice instead of using search filters — a real request CHS follows up on personally.	Bottom nav → Property request — Buyer / Tenant	Admin → Concierge, within Owner/Buyer/Tenant domain
Urgent & Emergency Sale browsing	Real, discounted, time-boxed listings shown in a dedicated homepage row with the real deadline.	Homepage — Buyer	N/A — browsing only

Feature	How it works	User location	Admin location
Market data (average prices, demand trend)	Real, genuine data (not placeholder figures) on current prices and search-driven demand patterns.	Bottom nav → Average prices / Demand trend	N/A — self-service
Engage CHS — buy a construction project	Full request flow with real specifications, document checklist, two-way chat, and the Construction Roadmap tool.	Engage CHS page — Buyer / Owner	Admin → Engage CHS (Super Admin only)
Hotel, Event Centre, Hall, Car Park, Cinema & Recreational/Sports booking	A real, instant "Book now" flow — not a rental application — for any genuine guest venue, correctly categorized and commissioned. A business-lease "hire" listing (e.g. a filling station) correctly stays on the rental-application path instead.	Property detail page — Buyer / Guest	N/A — self-service
Auto-pay rent from a dedicated savings wallet	An optional, real toggle — rent pays itself the moment it's due from a separate Rent Savings balance, or stays fully manual if left off.	Tenant dashboard, per tenancy	N/A — self-service
Subsequent-year rent — no repeat tenant commission	The system now genuinely knows a renewal from a first payment. From year two onward, the tenant pays CHS nothing further — only the landlord, at a real, reduced 3%. Shown live on both the tenant's and owner's rent countdown.	Tenant & Owner dashboards, rent countdown	N/A — automatic
Move-out condition report + real court affidavit	A genuine move-out counterpart to the existing move-in report, with a real, required court-issued affidavit upload — now actually visible to the landlord, which it never was before.	Condition Report page — Tenant; visible on Owner dashboard	N/A — self-service
Non-renewal notice with a real message	A genuine optional message sent to the landlord along with the notice — previously a bare, wordless click.	Tenant dashboard, per tenancy	N/A — self-service
Real street address on every listing	A required, specific address (e.g. "House 12, Shagari Road") — not just a general neighborhood — visible on the tenant's own dashboard and usable for real fault/payment tracing.	List a property form; Tenant dashboard	Admin analytics, traceable by real address

3. Property Owner

Feature	How it works	User location	Admin location
List a property	Photos, location, price, purpose, bedrooms, and special terms (rent-to-own, house rules). Starts unverified.	Owner dashboard → List a property	Admin → Properties (Owner/Buyer/Tenant domain)
Property verification	CHS reviews every listing before it's publicly visible — the real meaning behind "CHS Verified."	Owner dashboard	Admin → Properties (Owner/Buyer/Tenant domain)
Fixed-tier promotion (7/30/90-Day Boost)	One-off paid tier, debited directly from the wallet.	Promote page, per listing	N/A — self-service
Credit-based promotion (pause anytime)	Real, shared credit balance across every listing; each listing charged separately per real day it's on, based on real location/size cost. Real ranking (Category A–D + literal Top 10/20 position) against the same real local market.	Promote page, per listing	N/A — automatic, database-driven
Urgent & Emergency Sale activation	Requires the listing already verified and owner ID-verified. Real original price, discount, and deadline shown to buyers.	Owner dashboard → 🚨 Urgent Sale	Notified via general notifications on activation
Managing offers & rental applications	Real accept/reject decisions on real, CHS-screened offers and applications.	Owner dashboard	Admin → Sale Approvals / Applications
Inspection management + Readiness Score visibility	See each real requester's readiness score; mark real attendance/no-show after the scheduled time passes.	Owner dashboard, per property	Admin → Inspections
Delegating to a Property Manager	Hands maintenance and certain approvals to a registered Manager; reversible any time.	Owner dashboard	N/A — self-service
Property analytics	Real view counts, inspection counts, and interest counts — not fixed placeholder numbers.	Analytics page, per listing	N/A — self-service
Editing a live listing	Real edits to an already-published property.	Edit listing page	N/A — self-service
Engage CHS for a professional service	Full Property Management, Sale Negotiation, Construction Monitoring, Project Management, Renovation — each with its own real terms and fee schedule.	Engage CHS page — Owner	Admin → Engage CHS (Super Admin only)
Download verified sale documents	The actual, CHS-verified legal documents (Certificate of Occupancy, Deed of Assignment, etc.) genuinely downloadable by a paying buyer directly — previously only ever tracked as a physical-dispatch status with no real digital access.	Property detail page, after payment — Buyer	N/A — self-service
Accept an offer at negotiated terms, mid-chat	A real conversion point that didn't exist before — accept at a different final price than the original offer, with an optional condition (e.g. a payment deadline), directly from the ongoing negotiation.	Owner dashboard, per offer	N/A — self-service

4. Agent

Feature	How it works	User location	Admin location
Unique referral link	A real code (e.g. CHS-AG-0024) automatically attached to every property link generated — always opens the property directly on CHS.	Agent dashboard	N/A — self-service
Automatic commission attribution	If a referred visitor completes a real transaction on that property, CHS automatically attributes and credits the commission.	Agent dashboard	Admin → Referral fees (Agent Relations domain)
Real referral tracking (masked for privacy)	Real deal stage, earnings this month, and total earned — served from a privacy-conscious masked view.	Agent dashboard	Admin → Referral fees
Managing an agent-managed property's own commission	For a property under full agent authority, set your own real market rate (e.g. 10-15%) with the client — CHS charges the client nothing extra, taking a real, capped 3% only from your own earnings.	Agent dashboard, per managed property	N/A — self-service
Per-owner commission rate — one real, standing rate	A genuine, editable rate per real owner relationship, not re-entered per property — reviewable and adjustable any time, applying automatically to every property you manage for that owner, including new ones.	Agent dashboard → Your Real Commission Rate Per Owner	N/A — self-service
Digital remittance to an owner's wallet	Rent collected in person, remitted straight into the real owner's wallet with your agreed commission auto-deducted — a real, permanent reference either side can point to.	Agent dashboard → Remit Collected Rent to an Owner	Admin → Finance, real remittances table
A real tenant register	Full biodata, property details, and a real auto-generated reference number (e.g. TEN-000001) for every tenant you manage — plus real ID and selfie verification.	My Tenant Register page	N/A — self-service
A real mini-admin — your own team	Add real, separate staff accounts by phone number, assign a role, and see their genuine daily activity reports — the same kind of oversight CHS itself runs on.	Agent dashboard → Manage My Team	N/A — self-service
Multiple linked bank accounts	Up to 4 real, Paystack-verified bank accounts — a genuine business need for agents working across multiple banking relationships, unlike every other role which stays single-account.	Profile → Linked bank accounts	N/A — self-service
Expenses & income tracking	Real salary, rent, transport, logistics, and commission income entries — a genuine net position, not just a spending log.	My Expenses & Income page	N/A — self-service
My real activity report	A genuine, date-range (week/month/quarter) report of commission earned, deals closed, and tenancies managed.	Agent dashboard → Generate my activity report	N/A — self-service

5. Property Manager

Feature	How it works	User location	Admin location
Delegated property management	Owners can delegate specific properties; maintenance and certain approvals route to the manager directly.	Manager dashboard	N/A — self-service
Management termination	An owner can reverse delegation any time, handing responsibilities back.	Owner dashboard (trigger) / Manager dashboard (effect)	N/A — self-service
Posting a maintenance job for quotation	Real job posts that verified artisans can quote on, ranked by a transparent rating/reliability/experience/equipment formula.	Manager dashboard	Admin → Maintenance (Artisan/Developer/PM/Vendor domain)
Rating an artisan	Real, genuine ratings after a completed job.	Manager dashboard, per completed job	N/A — self-service
Owner-occupier support in estates	Register a real occupant who owns their unit outright (no tenancy) alongside genuine tenants in the same estate — correctly counted in occupancy and included in service-charge billing.	Manager dashboard, per estate unit	N/A — self-service
Per-owner commission rate, digital remittance, tenant register, mini-admin team, multiple bank accounts, expenses tracking	Every real business tool built for agents this round applies identically to managers — the same standing per-owner rate, the same digital remittance, the same tenant register and team system.	Manager dashboard — same real pages as Agent	N/A — self-service
My real activity report — across all my estates	A genuine, date-range report of new tenancies, service charges billed and collected, and maintenance resolved across every estate you manage.	My Estates page → Generate my activity report	N/A — self-service

6. Commercial Developer

Feature	How it works	User location	Admin location
Developer registration & review	A real application reviewed by CHS before projects go live.	Register / Developer dashboard	Admin → Developers (Artisan/Developer/PM/ Vendor domain)
Listing development projects	Estates, instalment plans, and investment plans.	Developer dashboard	Admin → Developers

7. Vendor & Marketplace

Feature	How it works	User location	Admin location
Vendor registration & verification	Reviewed by CHS before listings go live — the same trust standard applied platform-wide.	Become a Vendor / Vendor dashboard	Admin → Vendors (Artisan/Developer/PM/ Vendor domain)
Listing products & bundles	Real materials/services catalog, browsable by category on the Marketplace.	Vendor dashboard (list) + Marketplace page (browse)	N/A — self-service
Service quote requests	Real incoming requests for a quote on a specific service.	Vendor dashboard	N/A — self-service
Referral fee tracking	Real fees on deals connected through the Marketplace.	Vendor dashboard	Admin → Referral fees

8. Artisan

Feature	How it works	User location	Admin location
Artisan registration & verification	Real trades, years of experience, certification, and equipment tier (with photo/receipt backing) — reviewed before quoting is allowed.	Become an Artisan / Artisan dashboard	Admin → Artisans (Artisan/Developer/PM/ Vendor domain)
Quoting on maintenance jobs	Real quotes ranked by a transparent formula: rating/reliability first, experience second, equipment third.	Artisan dashboard	Admin → Maintenance
Full-management job priority	For any property under full CHS management, maintenance work goes first and exclusively to verified CHS Maintenance Agents.	Automatic — Artisan	N/A — automatic
Job dispute resolution	Either the artisan or the client can raise a genuine, two-sided dispute about a completed job.	Artisan dashboard	Admin → Disputes
Ratings received	Real, visible rating history.	Artisan dashboard	N/A — self-service

9. Admin & Platform Management

Feature	How it works	User location	Admin location
Super Admin role	One real account, practically involved in every part of the platform — handles Finance and Engage CHS directly, and ratifies every sub-admin's high-stakes decision.	Admin dashboard	Admin → Overview
Five real sub-admin domains	Customer Care, Registration & Setup, Owner/Buyer/Tenant, Agent Relations, Artisan/Developer/PM/Vendor — each sub-admin only sees and acts within their own domain, enforced at the database level.	Admin dashboard, tab-gated per domain	Admin → Overview → Assign an admin role
Code-based admin login approval	A sub-admin's correct password alone isn't enough — a real code is generated and sent to the Super Admin, who must approve that specific login.	Sub-admin login flow	Admin → Overview — pending login approvals
High-stakes approval queue	Routine sub-admin actions apply immediately; financial and verification decisions are held pending real Super Admin sign-off.	Admin dashboard, across relevant tabs	Admin → Overview — pending sub-admin actions
Trace an Account	Search any real user by phone/email/name and see their full wallet balance + transaction history, promotion credits + active promotions, construction roadmap unlocks, linked bank account, and Engage CHS requests — all in one place. The real fix for resolving a support issue ("I paid but nothing happened") the way a phone network traces an account from just a phone number.	N/A — admin only	Admin →  Trace an Account (Super Admin only)
Sub-admin action history	A real, permanent audit log of every resolved sub-admin request — previously, once approved or rejected, a request simply vanished with no record.	N/A — admin only	Admin → Overview → View sub-admin action history
Pre-launch admin test mode	Super Admin can preview any real dashboard using their own account, never a real user's data, with a visible banner the whole time.	N/A — testing tool	Admin → Overview → Switch role for testing
Finance	Search any real user's wallet, view all three real balances, freeze/unfreeze.	N/A — admin only	Admin → Finance
Sale Approvals	The real checkpoint between an accepted offer and money moving to escrow.	N/A — admin only	Admin → Sale Approvals
Face Verification (Liveness)	Real human review of each real-time capture.	N/A — admin only	Admin → Face Verification
Registrations	Approve or reject new real accounts.	N/A — admin only	Admin → Registrations
Applications	Real rental application review before an owner ever sees it.	N/A — admin only	Admin → Applications
Properties	Real listing verification before it's publicly visible.	N/A — admin only	Admin → Properties
Disputes	Real, two-sided dispute resolution across tenancies and artisan jobs.	N/A — admin only	Admin → Disputes
Feedback	General platform suggestions/complaints, not tied to a specific transaction.	N/A — admin only	Admin → Feedback
Engage CHS	Real professional-service requests: specifications, document checklist, two-way	N/A — admin only	Admin → Engage CHS

Feature	How it works	User location	Admin location
	chat, roadmap access — Super Admin only, per direct instruction.		
Vendors	Real marketplace vendor verification.	N/A — admin only	Admin → Vendors
Referral fees	Real agent/vendor commission tracking and payout, routed through the approval queue.	N/A — admin only	Admin → Referral fees
Maintenance	Real fault report routing to the right artisan/manager.	N/A — admin only	Admin → Maintenance
Artisans	Real trade/equipment verification before an artisan can quote.	N/A — admin only	Admin → Artisans
Inspections	Real upcoming inspection oversight across the whole platform.	N/A — admin only	Admin → Inspections
Developers	Real developer application review.	N/A — admin only	Admin → Developers
Real, date-range analytics report	A genuine, filterable (today/week/month/quarter) record of properties sold, new tenancies, shortlet bookings, new listings, new users, and total commission revenue by transaction type — not just a live, un-datable snapshot.	N/A — admin only	Admin → Analytics
Suspend a real account, with a genuine appeal process	Suspend by phone number with a required, documented reason; a real appeal queue to approve (reinstate) or deny.	N/A — admin only	Admin → Overview
CHS staff daily report	Any real internal admin staff member can submit their own daily activity report, visible to the super admin.	N/A — admin only	Admin → Overview → Submit My Daily Report
Critical data-protection fixes	A direct security audit found and closed a real, proven exposure — any logged-in user could previously read another user's NIN, phone, and ID documents. Fixed at both the database and file-storage level, re-verified after the fix.	N/A — platform-wide	See the separate Security Audit report
Missing database indexes added	Dozens of real foreign-key columns had no index — invisible at small scale, a genuine slowdown waiting to happen as real data grows. Added across every high-traffic table.	N/A — platform-wide	See the separate Scalability Assessment report
Real, editable Contact Us details	Four real, distinct contact emails (Support, Inquiry, Engage CHS, Admin) and two real phone numbers — genuinely admin-editable at any time, not hardcoded into the app.	More menu → Contact us — every role	Admin → Overview → Edit Real Contact Details

10. Estate Management, Rent to

Feature	How it works	User location	Admin location
Estate Management	A real, bounded estate of many housing units under one manager, distinct from delegating individual properties one at a time.	Manager → My Estates	Admin → Overview
Estate subscription	Real monthly pricing that scales with unit count, and drops per-unit as an estate grows — paid directly from the manager's wallet.	Manager → My Estates → Subscribe	N/A
Bulk unit onboarding	Upload a real CSV to add many units at once instead of one at a time — the real answer to onboarding an estate of hundreds of units.	Manager → My Estates → [Estate]	N/A
Estate service charges	A real, distinct-from-rent charge billed to every occupied unit at once; vacant units are automatically skipped.	Manager dashboard / Tenant dashboard	N/A
Rent to Own / Mortgage	A genuinely new listing category — progressive ownership through real monthly installments, with real ownership-percentage tracking that automatically completes the sale at 100%.	Property listing page / Owner dashboard	N/A
Reminder Engine	One real, generic escalation system: rent-due cadence, maintenance follow-up (artisan → manager), and shortlet guest-message escalation (host → admin) — multi-channel (in-app, SMS, email).	N/A — automatic, background	N/A
Shortlet/Hotel guest-host messaging	Real in-app communication for the duration of a booking, with genuine host anonymity toward the guest — never toward CHS, which can always intervene directly.	My Bookings (guest) / Owner dashboard (host)	N/A
Maintenance Reserve — real funding	An owner can fund their Maintenance Reserve directly from their Main Wallet for a real, larger repair; artisan payments draw from the Reserve first, and unused funds can be withdrawn back freely.	Wallet page	N/A

11. Real Sale Payment, Escrow,

Feature	How it works	User location	Admin location
Real Sale purchase-price payment	The genuine mechanism for a buyer to pay a seller the actual purchase price — not just commission. A single, transparent checkout: the buyer's total automatically includes their own commission (shown as both a percentage and a real value), the seller's net automatically excludes theirs.	Property page, once an offer is accepted	N/A
Real, sourced Sale legal documents	Certificate of Occupancy, Deed of Assignment, Survey Plan, Governor's Consent, Tax Clearance Certificate, Sale Agreement, and (for a developed property) Building Plan Approval — sourced from genuine Nigerian legal references. Required at listing time; a buyer's payment is blocked until every one is CHS-verified.	List a property (Sale)	Admin → Face Verification tab (Sale documents)
Escrow hold on sale proceeds	A seller's real proceeds are visible in their wallet the instant a buyer pays, but not withdrawable until CHS confirms the real, physical legal documents have been transferred to the buyer.	Wallet page	Admin → Registrations tab (Confirm legal transfer)
Real, admin-reviewed identity verification	A buyer's ID submission previously self-certified the instant the form was filled in. Now creates a genuine pending record, reviewed and approved by real CHS staff before it counts.	Making an offer (if not yet verified)	Admin → Registrations tab
Mobile logout	A working logout in the persistent bottom navigation's More menu — previously only reachable from the homepage.	Bottom nav → More	N/A
Context-aware House Rules	Only appears for Rent, Lease, Shortlet, and Rent to Own listings — no longer shown on a Sale listing where it never applied.	Owner dashboard	N/A
Real, progressive nearby-properties matching	The "similar properties" advertising system now genuinely widens from exact area → LGA → state, and matches property type, rather than matching the whole state regardless of relevance.	Property detail page	N/A
Seller response note on an offer	A real, optional message a seller can attach when accepting or declining an offer — delivered directly to the buyer's notification, matching the note a buyer could already leave.	Owner dashboard	N/A
Rent countdown and 90-day non-renewal notice	A real, live day-count to next rent on the tenant dashboard, and a genuine notice mechanism that honestly records whether notice was given within or after the requested 90-day window.	Tenant dashboard	N/A

12. Round 3 Client Testing — Re

Feature	How it works	User location	Admin location
Verification review moved to the right place	A real, repeated complaint — buyer ID verification submissions "vanished." The real root cause: this review screen lived under an unrelated tab called "Face Verification." Moved into "Registrations," where it's actually found.	N/A — admin only	Admin → Registrations tab
Gender field and a respectful greeting	A real gender field at registration — the homepage now greets "Hi Mr. Samson" / "Hi Miss Jennifer" instead of a bare first name.	Register page; Homepage	N/A — self-service
Dynamic, per-role comprehension questionnaire	Real, separate question banks for Buyer, Guest, Tenant, Owner, Agent, and Manager — each genuinely testing that role's own real Terms & Conditions clauses, not one static quiz for everyone.	Register page	N/A — self-service
Required-field markers and auto-scroll-to-error	A real red asterisk on every genuinely required field; submitting an incomplete form now scrolls straight to the exact missing field and highlights it in red.	Register page	N/A — self-service
Real State → LGA multi-select for agent coverage area	A genuine checklist of every real LGA in the agent's state, plus a "Statewide" shortcut that selects them all in one click.	Register page (Agent)	N/A — self-service
"Others" registration category for artisans, market sellers, and service providers	A real, new registration option leading to a genuine "choose your category" page, bridging straight into the existing Artisan and Vendor application forms that were previously unreachable from registration.	Register page; Login page	N/A — self-service
Real login for Artisans and Vendors/Service Providers	Confirmed and fixed a genuine gap: someone approved as an Artisan or Vendor had no way to log back in as themselves. Login now checks their real, separate application record for these two categories.	Login page	N/A — self-service
CHS-specific inspection-fee question	Reworded to specifically ask about a CHS agent acting on CHS's own commission structure — CHS cannot control what an independent agent charges their own client.	Register page	N/A
Independent agent fee builder and real Total Package	An agent listing a property can add any number of named, real fee lines (Custom fee, Legal fee, Agent fee, etc.) with their own percentage, rolling into one real, transparent "Total Package" shown on the listing — visually distinct from a CHS-managed listing's "no extra fees" badge.	List a property (Agent); Property detail page	N/A — self-service
Real, live admin notification bell	A genuine, real-time notification bell with an unread badge — database-level triggers now automatically notify admin the instant a new registration or ID verification comes in, regardless of which part of the app created it.	N/A — admin only	Admin dashboard header
Real shortlet booking data fix	Found and fixed the actual cause of shortlet bookings failing outright — 20 of 22 real listings were missing their nightly price. Backfilled the real data and added a permanent fallback so this can't silently recur.	Property detail page (Guest)	N/A

Feature	How it works	User location	Admin location
Correct role-based landing after login/registration	Fixed two real, separate bugs sending every role to the generic homepage — the registration redirect, and a completely separate Face ID/fingerprint login path that ignored role entirely.	Login page; Register page	N/A
Staff login, made discoverable and functional	Confirmed the Manage-My-Team and staff-reporting features already existed; found and fixed the real, missing piece — a brand-new staff member had no role to register under. "Staff / Employee" is now a real registration option.	Register page; Login page; /staff	N/A — self-service
Real "My Properties & Owners Register"	A genuine register of every property and owner under an agent/manager's real management — property, address, owner name, owner phone, and agreed commission rate — parallel to the existing tenant register.	Agent/Manager dashboard	N/A — self-service
Real one-step-back navigation	Fixed a hardcoded "back to homepage" link repeated across 15 real sub-pages — each now genuinely returns one real step, not straight to the homepage.	Every sub-page — every role	N/A
Detailed, consequential guarantor form	Added real relationship-to-applicant, guarantor address, guarantor occupation, and a genuine, plain-language consent statement — not just a bare name and phone number.	Rental application (Tenant)	N/A — self-service
Real, complete rental application review for admin	Admin can now see full applicant details (occupation, address, income, ID, uploaded document) and full guarantor details, plus which real property and owner it's for — genuine vetting is now possible before forwarding.	N/A — admin only	Admin → Applications tab
CHS-mediated owner decision on rental applications	An owner's approve/decline is now recorded, not finalized immediately — CHS admin reviews and relays the real, final decision to the applicant, keeping CHS genuinely in the loop both ways.	Owner dashboard; Admin → Applications tab	Admin → Applications tab

13. AgentManager Tools, HostG

Feature	How it works	User location	Admin location
Real team/staff system	An agent or manager can invite genuine, separate staff accounts, assign a role label, and see each one's real daily activity report.	Agent/Manager dashboard → Manage My Team	N/A — self-service
Team/staff subscription	Free for up to 2 real staff; a real subscription required from the 3rd, tiered by staff count, with exact multi-month discounts (6 months paid → 8 months of real access; 12 paid → 18).	Agent/Manager dashboard → Manage My Team	N/A
Real tenant register and property/owner register	Per-agent, searchable registers with real ID documents attached, replacing an informal spreadsheet.	Agent/Manager dashboard	Admin → Tenant Register Oversight
Expense tracking, owner remittance, digital receipts	Real income/expense entries, a real flow for remitting collected rent to the actual owner, and auto-generated digital receipts for completed transactions.	Agent/Manager dashboard → Expenses	N/A
Independent agent commission model and custom fee builder	A real, adjustable per-owner commission rate, plus any number of named custom fee lines rolling into one transparent Total Package.	Agent dashboard; List a property (Agent)	N/A — self-service
Host and Guest as real, separate roles	Split from Owner/Buyer with their own dedicated dashboards, built the same way as every other role.	/host; /guest	N/A — self-service
Real shortlet/hire request-to-book flow	Complete rebuild: a guest sees the true, full cost (rent plus commission) before committing; a host must genuinely accept or decline; a decline triggers a real, automatic full refund. Previously, bookings were instantly confirmed with no host review and no real way for a host to ever get paid.	Property page (Shortlet/Hire)	N/A
Real host payout	Released deliberately by admin, net of the host's own commission — the same CHS-mediated pattern used for sale-proceeds release.	N/A — admin only	Admin → Shortlet/Hire Deposits & payouts
Two-way ratings	Guest and host can each rate the other once, after a confirmed stay.	Guest dashboard; Host dashboard	N/A
Real, stated cancellation policy	Server-enforced: full refund 48+ hours before check-in, 50% within 48 hours, none after.	Guest dashboard	N/A
Flexible security deposit	Host-toggled per listing; applies only to a genuinely first-time guest and auto-waives after their 3rd real rating. Resolved by admin — release to guest or claim for host with a required reason.	List a property (Shortlet/Hire)	Admin → Shortlet/Hire Deposits
House-rules attestation	A guest must now see and check a box agreeing to a property's real house rules before requesting to book — closing a gap where the feature existed for hosts but was invisible to guests.	Shortlet/Hire booking form	N/A
Dispute-filing for Guest and Host	A working dispute form already existed for Tenant/Owner; now genuinely reachable from Guest and Host dashboards too.	Guest dashboard; Host dashboard	Admin → Disputes tab
Developer as a real,	Could previously register but had no login	/developer	N/A — self-service

Feature	How it works	User location	Admin location
separate role	path and no dashboard at all. Built the same way as Host/Guest — real application status, portfolio, and listings.		
Real ID-number field per document type	Fixed a genuine bug: choosing any ID type other than NIN had nowhere to enter that document's own number — the system was silently saving the person's NIN instead. A real, conditional number field now appears per ID type.	Register page	N/A — self-service
Multi-role KYC review fix	Admin's document review only ever checked a person's primary role, missing real credentials attached to a secondary role — found and fixed against a real, in-house test case.	N/A — admin only	Admin → Registrations tab
Fresh, live document links	Document links previously relied on a URL signed once and stored permanently — a real structural fragility. Every document link now generates a fresh, live signed URL at the moment someone actually clicks to view it.	Throughout admin and owner/host document review	N/A
Property reference numbers and admin search	Every property now has a real, permanent reference number (PROP-000001 style), backfilled for every existing property. A real admin search finds any property by reference, title, or the real owner's name/phone.	N/A — admin only	Admin → Properties tab
Real PIN reset	Phone + NIN verification, since the platform has no email/SMS infrastructure. Did not exist in any form before this.	Login page → Forgot your PIN?	N/A — self-service
Real FAQ page	Written from CHS's own actual Terms & Conditions and features, not copied from another platform.	More menu → Help & FAQs	N/A
Add a role without logging out, plus a role switcher	An already-authenticated session now skips the phone/PIN re-verification entirely when adding a role. Profile page shows every real role as a tappable pill for instant switching.	Profile page → Add another role	N/A — self-service
Desktop splash screen fix	Reproduced directly with a real screenshot before fixing — a background image built for a phone's tall aspect ratio was being cropped into an unrecognizable dark sliver on a wide desktop screen. Mobile confirmed untouched.	Login page (first load)	N/A
Four new per-role color themes	Host, Guest, Developer, and Staff — previously quietly borrowing another role's colors — now each have their own genuinely distinct color identity.	/host; /guest; /developer; /staff	N/A

14. Marketplace Rebuild, Guaran

Feature	How it works	User location	Admin location
Redesigned receipt and payment voucher	A real, branded document — gradient header, a genuine watermark seal, proper serif typography — replacing a plain data table, per direct client feedback.	/receipt/[reference]	N/A
Real, tiered artisan commission	6% for labor under ₦100,000, 8.5% at or above, on labor only — never on materials procured separately. Replaces an earlier flat 10%.	N/A — automatic on job payment	N/A
Marketplace: admin-mediated, contact-filtered messaging	Every quote request and vendor response is held for real CHS review — a phone number or email in either direction is caught and blocked automatically — before the other party ever sees it.	Marketplace → Request a quote	Admin → Marketplace Moderation
Marketplace: real, escrow-backed payment	Accepting an approved quote charges the buyer the real price plus 6%, held in escrow; the vendor receives the price net of 4% only once CHS confirms delivery. Full refund available if a deal falls through.	My Requests → Accept & Pay	Admin → Marketplace Moderation
Marketplace: real direct "Buy Now"	Sort by price and buy the cheapest listing directly at its own fixed price — no negotiation, no messages to moderate. Same real 6%/4% split and escrow protection.	Marketplace → Buy now	Admin → Marketplace Moderation
Sister marketplace link (Unify Market Central)	A real, admin-editable link for procurement CHS's own marketplace doesn't cover — still real, verified sellers, still under CHS's own jurisdiction.	Marketplace; Construction Roadmap; Owner → Maintenance	Admin → Overview → Edit Real Contact Details
Rental application: real applicant detail shown to owner	The owner's review card previously showed only the guarantor's name, phone, and move-in date. Now shows the applicant's real name, ID-verified status, occupation, income, employer, and current address — every field already being collected, now actually displayed.	Owner dashboard → Rental applications	N/A
Independent guarantor verification	The applicant provides only the guarantor's name and phone. The guarantor completes every other real field about themselves — address, occupation, relationship, their own ID, and their own typed-name consent — via a private, single-use link, requiring no CHS account. The application cannot reach the owner until this is done.	New applicant flow → real link sent to guarantor → /guarantor-confirm/[token]	N/A
Event Centre: real capacity tiers and priced facilities	Real, host-set price tiers by guest count, and a real priced facilities list (lighting, sound, catering per guest, live band, DJ, decoration, generator, security, parking attendants, ushers/event staff) with a live-calculated quotation as the guest selects.	Event Centre booking form	Host → Real Event Pricing (/event-pricing/[propertyId])
Event-day add-on requests	A real request for music band, caterer, and/or ushers (with a count) on any hire/event booking, visible directly on the host's own dashboard.	Hire/Event booking form	N/A
Real payment regression fixes —	Both previously charged only the base amount, silently omitting the payer's own real,	Tenant dashboard → Pay rent; Rent-to-Own	N/A

Feature	How it works	User location	Admin location
rent and Rent-to-Own	already-invoiced commission. Both now compute and charge the true total in one real payment, shown to the payer beforehand.	installment payment	
Stale, uncollected commission alert	A real, visible warning on Admin → Finance for any commission invoiced but genuinely uncollected for 2+ hours — an early warning against the exact regression pattern found and fixed above.	N/A — admin only	Admin → Finance
Clickable notifications with real navigation	A rental application's full decision chain — owner decides, CHS relays, tenant is told — now carries a real, working link at every step, straight to the relevant page.	Notification bell, any role	Notification bell
Real-time notification delivery	The notifications table is now genuinely part of Supabase's real-time publication — confirmed missing before, and the actual cause of alerts never appearing without a manual refresh. A real, visible toast now appears the instant a new notification arrives.	Any dashboard, any role	Any dashboard
My Applications	Every real rental application and purchase offer a buyer/tenant has made, queued permanently the moment they apply — no more scrolling the homepage to find it again.	/my-applications	N/A — self-service
Recent Applications (owner)	Every real application received across every property, most recent first, regardless of how far down the property list it sits.	Owner dashboard → Recent Applications	N/A — self-service
Platform Earnings	A real, dedicated view of every collected commission — payer name, amount, and timestamp.	N/A — admin only	Admin → Platform Earnings
Rent Savings relocated	Confirmed the tenant dashboard already showed this correctly; removed the redundant duplicate from the general homepage.	Tenant dashboard	N/A